

Jeonghyun Woo

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💡 RESEARCH INTERESTS

My research advances **memory systems**, **computer architecture**, **hardware security**, and **AI/ML systems**, focusing on:

- **Secure and Efficient Memory Systems:** Building high-performance, energy-efficient, and secure memory systems across conventional and emerging technologies, with defenses against RowHammer, denial-of-service, and side-channel attacks.
- **Efficient LLM Inference:** Co-designing algorithms and memory systems to reduce data movement and latency in large-scale LLM serving, including efficient KV-cache management and speculative decoding.
- **Trustworthy AI Systems:** Designing lightweight cross-layer defenses against hardware-induced privacy and integrity threats in LLM serving, including bit-flip jailbreaks and side-channel leakage from MoE and speculative decoding.

🎓 EDUCATION

The University of British Columbia (UBC)

Ph.D., Electrical and Computer Engineering | GPA: 4.00/4.00

Advisor: Prof. Prashant Nair

Dissertation: *Secure and Low-Overhead RowHammer Mitigations for Scalable Memory Systems*

September 2022 – Present

Vancouver, BC, Canada

Hanyang University (HYU)

M.S., Electronics and Computer Engineering | GPA: 4.00/4.00

Advisor: Prof. Ki-Seok Chung

Thesis: *RowHammer Mitigation Architecture for Highly Reliable DRAM*

March 2018 – February 2020

Seoul, South Korea

Hanyang University (HYU)

B.S., Electronic Engineering, Summa Cum Laude | GPA: 3.89/4.00

March 2012 – February 2018

Seoul, South Korea

💼 PROFESSIONAL EXPERIENCE

The University of British Columbia (UBC)

Graduate Research Assistant, **STAR Lab**

Advisor: Prof. Prashant Nair

September 2022 – Present

Vancouver, BC, Canada

NVIDIA Research

Research Intern, **Architecture Research Group**

Mentor: Dr. Aamer Jaleel | Manager: Dr. David Nellans

May 2024 – August 2024

Westford, MA, USA

Micron Technology

Systems Research Engineering Intern, **Vertical Systems Research**

Mentor: Dr. Chun-Yi Liu | Manager: Ameen Akel

May 2023 – August 2023

Folsom, CA, USA

University of Illinois Urbana-Champaign (UIUC)

Graduate Research Assistant, **Systems Platform Research Group**

Advisor: Prof. Jian Huang

August 2020 – January 2021

Champaign, IL, USA

Hanyang University (HYU)

Graduate Research Assistant, **ESoC Lab**

Advisor: Prof. Ki-Seok Chung

March 2018 – February 2020

Seoul, South Korea

🏆 HONORS AND AWARDS

Distinguished Artifact Award, HPCA 2025 — 1 of 2 Distinguished Artifacts 2025

Best Paper Award, HPCA 2023 — 1 of 2 Best Papers from 364 submissions 2023

Student Travel Grants — ISCA (2025, 2026); HPCA (2023, 2025) 2023, 2025, 2026

Faculty of Applied Science Graduate Award, UBC — \$24,800 CAD total 2022, 2023, 2024, 2025

Hanyang Graduate School Scholarship — 70% tuition coverage (~\$9,200 CAD/yr) 2018, 2019

Outstanding Paper Award, KICS Fall Conference 2017

Hanyang Academic Excellence Award — Top 1% of ~15,000 university students 2016, 2017

Hanyang Alumni Association Scholarship — Full tuition (~\$10,000 CAD/yr) 2016, 2017

Excellent Tutor Award, Engineering Mathematics Tutoring Program, Hanyang University 2016

Hanyang University Scholarship — Full tuition (~\$10,000 CAD/yr) 2012, 2013

Peer-Reviewed Conference Publications

- [C6] “Loaded Dice: Solving the Non-Selection Problem for Scalable Probabilistic RowHammer Defense.”
Jeonghyun Woo*, Junsu Kim*, Aamer Jaleel, and Prashant J. Nair. (**equal contribution*)
53rd Annual International Symposium on Computer Architecture (ISCA’26), to appear. Acceptance Rate: 19.1%.
 Paper  Code
- [C5] “When Mitigations Backfire: Timing Channel Attacks and Defense for PRAC-Based RowHammer Mitigations.”
Jeonghyun Woo, Joyce Qu, Gururaj Saileshwar, and Prashant J. Nair.
52nd Annual International Symposium on Computer Architecture (ISCA’25), June 2025. Acceptance Rate: 23.1%.
 Paper  Code  Slides
[Artifact Evaluated:  Available,  Functional,  Reproduced]
- [C4] “QPRAC: Towards Secure and Practical PRAC-Based Rowhammer Mitigation Using Priority Queues.”
Jeonghyun Woo, Shaopeng (Chris) Lin, Prashant J. Nair, Aamer Jaleel, and Gururaj Saileshwar.
31st International Symposium on High-Performance Computer Architecture (HPCA’25), March 2025. Acceptance Rate: 21.0%.
 Paper  Code  Slides
[ Distinguished Artifact Award — 1 of 2 Distinguished Artifacts]
[Artifact Evaluated:  Available,  Functional,  Reproduced]
- [C3] “DAPPER: A Performance-Attack-Resilient Tracker for RowHammer Defense.”
Jeonghyun Woo and Prashant J. Nair.
31st International Symposium on High-Performance Computer Architecture (HPCA’25), March 2025. Acceptance Rate: 21.0%.
 Paper  Slides
- [C2] “Scalable and Secure Row-Swap: Efficient and Safe Row Hammer Mitigation in Memory Systems.”
Jeonghyun Woo, Gururaj Saileshwar, and Prashant J. Nair.
29th International Symposium on High-Performance Computer Architecture (HPCA’23), February 2023. Acceptance Rate: 25.0%.
 Paper  Code  Slides
[ Best Paper Award — 1 of 2 Best Papers from 364 submissions]
[Artifact Evaluated:  Available,  Functional,  Reproduced]
- [C1] “HammerFilter: Robust Protection and Low Hardware Overhead Method for RowHammer.”
Kwangrae Kim, **Jeonghyun Woo**, Junsu Kim, and Ki-Seok Chung.
39th International Conference on Computer Design (ICCD’21), October 2021. Acceptance Rate: 24.4%.
 Paper  Slides  Video

Preprints and Under Review

- [P1] “PrisonBreak: Jailbreaking Large Language Models with at Most Twenty-Five Targeted Bit-flips.”
Zachary Coalson, **Jeonghyun Woo**, Chris S. Lin, Joyce Qu, Yu Sun, Shiyang Chen, Lishan Yang, Gururaj Saileshwar, Prashant J. Nair, Bo Fang, and Sanghyun Hong.
Submitted to *40th Annual Conference on Neural Information Processing Systems (NeurIPS’26)*, 2026.
 arXiv

Workshop Publications and Posters

- [W3] “CnC-PRAC: Coalesce, Not Cache, Per-Row Activation Counts for an Efficient in-DRAM Rowhammer Mitigation.”
Chris S. Lin, **Jeonghyun Woo**, Prashant J. Nair, and Gururaj Saileshwar.
Fifth Workshop on DRAM Security (DRAMSec’25), co-located with ISCA 2025, June 2025.
 Paper
- [W2] “Counterpoint: One-Hot Counting for PRAC-Based RowHammer Mitigation.”
Shih-Lien Lu, **Jeonghyun Woo**, and Prashant J. Nair.
Fifth Workshop on DRAM Security (DRAMSec’25), co-located with ISCA 2025, June 2025.
 Paper
- [W1] “HammerFilter: Robust Protection and Low Hardware Overhead Method for Row-Hammering.”
Kwangrae Kim, Junsu Kim, **Jeonghyun Woo**, and Ki-Seok Chung.
Work-in-Progress Poster, *58th Design Automation Conference (DAC’21)*, December 2021.
 Poster

Domestic (Korean) Conference Publications

- [D2] "A Method to Find the Optimal Probability for Probability-Driven Additional Row Refresh to Prevent DRAM Row Hammering."
Jeonghyun Woo and Ki-Seok Chung.
Korean Institute of Communications and Information Sciences (KICS) Winter Conference, January 2019.
- [D1] "Implementation of an FPGA-Based CNN Accelerator Using SDSoC."
Changwoo Lee*, Jeonghyun Woo*, Sang-Soo Park, and Ki-Seok Chung. (**equal contribution*)
KICS Fall Conference, November 2017.
[Code](#) [★ 300+ [GitHub stars](#)]
[🏆 Outstanding Paper Award]

🎤 INVITED TALKS

Built on Broken Ground: Why Secure Computing Must Start at the Hardware

University of Waterloo, Waterloo, Canada, March 2026

Towards Secure and Scalable Memory Systems: From RowHammer Attacks to Hardware-Induced LLM Jailbreaks

KAIST CASYS Seminar, Remote, November 2025

Towards Secure and Low-Overhead PRAC-Based RowHammer Mitigations

NVIDIA Research (end-of-internship talk), Westford, USA, August 2024

RowHammer Mitigations for Future High-Bandwidth Memory

Micron Technology (end-of-internship talk), Folsom, USA, August 2023

When Mitigations Backfire: Timing Channel Attacks and Defense for PRAC-Based RowHammer Mitigations

ISCA'25, Tokyo, Japan, June 2025

QPRAC: Towards Secure and Practical PRAC-Based Rowhammer Mitigation Using Priority Queues

HPCA'25, Las Vegas, USA, March 2025

DAPPER: A Performance-Attack-Resilient Tracker for RowHammer Defense

HPCA'25, Las Vegas, USA, March 2025

Scalable and Secure Row-Swap: Efficient and Safe Row Hammer Mitigation in Memory Systems

HPCA'23, Montreal, Canada, February 2023

Implementation of an FPGA-Based CNN Accelerator Using SDSoC

KICS'17 Fall, Daegu, South Korea, November 2017

🎓 TEACHING EXPERIENCE

The University of British Columbia (UBC)

Vancouver, BC, Canada

Graduate Teaching Assistant

CPEN 411: Computer Architecture

2022, 2023, 2024

CPEN 312: Digital Systems and Microcomputers

2025

Hanyang University (HYU)

Seoul, South Korea

Graduate Teaching Assistant

ELE 3081: VLSI Engineering

2019

ITE 4003: SoC Design

2018

👥 MENTORING EXPERIENCE

Junsu Kim

January 2020 – Present

Ph.D. Student, The University of British Columbia (UBC)

Vancouver, BC, Canada

Project: Scalable and robust RowHammer mitigations → **Published at:** DAC'21 (WIP), ICCD'21, ISCA'26

Project: Efficient LLM inference via KV-cache management and speculative decoding

Kwangrae Kim

January 2019 – December 2021

Ph.D. Student, Hanyang University (HYU)

Seoul, South Korea

Project: Low-cost and robust probabilistic RowHammer mitigation → **Published at:** DAC'21 (WIP), ICCD'21

🔧 PROFESSIONAL SERVICE

Program Committee

- 2026 IEEE/ACM International Symposium on Microarchitecture (**MICRO'26**) — Student PC
- 2026 IEEE International Symposium on High-Performance Computer Architecture (**HPCA'26**) — Light PC

Artifact Evaluation Committee

- 2025 IEEE International Symposium on Workload Characterization (**IISWC'25**)

Invited Reviewer

- IEEE Transactions on Very Large Scale Integration Systems (**TVLSI**), 2025

Student Volunteer

- 2025 SPICE Workshop, co-located with MICRO'25
- 2024 IEEE International Symposium on Workload Characterization (**IISWC'24**)
- 2023 International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS'23**)

🔗 SELECTED ACADEMIC PROJECTS

Compression-Assisted Last-Level Cache Replacement

January 2023 – April 2023

Advanced Computer Architecture, UBC (Instructor: Prof. Mieszko Lis)

Showed that 72.7% of cache lines are compressible by ≥ 10 B; proposed a compression-assisted LLC replacement policy that improves hit rates with minimal metadata overhead.

Revisiting Address Translation on Intel Optane DC PMEM

September 2022 – December 2022

Graduate Operating Systems, UBC (Instructor: Prof. Margo Seltzer)

Quantified address-translation overheads on Optane DC PMEM with graph processing, HPC, and genomics workloads; demonstrated that Transparent Huge Pages mitigate 4 KB page bottlenecks.

Modified LeNet-5 Forward Pass in CUDA

November 2020 – December 2020

Applied Parallel Programming, UIUC

Implemented optimized convolutional forward passes using shared-memory tiling, constant memory, and loop unrolling; profiled with Nsight Systems and Nsight Compute.

⚙️ TECHNICAL SKILLS

Programming: C/C++, Python, CUDA, Verilog, Perl, Bash, Go, Assembly

Simulators: Ramulator, ChampSim, gem5, DRAMSim2, GPGPU-Sim, MGPUSim

Hardware Tools: Pin, SimPoint, Xilinx Vivado, Xilinx SDSoC, Intel Quartus